Supplement: moving-world methods and condition definitions

This supplement supplies the construction that sec. 17.7 defers here: exactly how the moving-sentinel world is built, how the three actions and the expected-free-energy policy move an agent, and how the *isolated*,

Untagged PDF derivative | HTML reader

Supplement: moving-world... (part 2)

communicating, and *EFE-guided* conditions differ.

Supplement: moving-world... (part 3)

It answers the mechanical question left open in the main section — what precisely is held fixed and what varies across the three conditions whose accuracies are contrasted there — so the reported binary-complement numbers can be read against their generative model rather than taken on trust.

Supplement: moving-world... (part 4)

The moving-world generative model is built by `build_moving_world`. A linear grid of 4 cells holds one binary hidden state — the half of the grid (left = state 0, right = state 1) that contains the threat.

Supplement: moving-world... (part 5)

The 2 sentinels start at evenly tiled positions $(i \cdot \lfloor n_{\text{positions}}/n_{\text{agents}} \rfloor)$ and each observe a half-open field-of-view window. With the default setup the two FOVs are disjoint, one per half.

Supplement: moving-world... (part 6)

Each agent's likelihood is a 2×2 matrix over outcomes (detected / not_detected) given the binary state, with a confident signed reading for the half the agent watches; the transition tensor encodes three deterministic control paths — stay, left (reflecting at cell 0), and right (reflecting at the last cell).

Supplement: moving-world... (part 7)

The hidden-state prior is uniform.

Supplement: moving-world... (part 8)

Action selection has two regimes. The random conditions draw each agent's move uniformly from the three controls. The EFE-guided condition uses `efe_policy_select`: for every candidate move it lands the agent at the deterministic next position, reconstructs the likelihood from that viewpoint,

Supplement: moving-world... (part 9)

and scores the move by the expected posterior entropy after one observation, $H = \sum_o P(o)\, H(P(s \mid o))$ — taking the entropy-minimizing (information-seeking) step.

Supplement: moving-world... (part 10)

We compare three conditions — *isolated* (random moves, no sharing), *communicating* (random moves plus a per-step log-linear-pool consensus), and *EFE-guided* (information-seeking moves plus the same per-step sharing) — over 960 trials of 6 steps each,

Supplement: moving-world... (part 11)

scoring the pooled consensus against the true state. All numerics are deterministic given the run seed.